

Introduction to Applied Artificial Intelligence

Dr. Huseyin Kusetogullari

Who am I?



PhD, 2012

Intelligent Systems Lab., School of Engineering, Warwick University, U.K.

CURRENT POSITION

Associate Professor (Docent), Department of Computer Science, Blekinge Institute of Technology, Sweden.

Program Manager in AI / ML, BTH

PROJECTS

- Road Surveillance Systems to provide safety and security for vehicles and pedestrians.
- Automated recognition of human subjects via gait recognition and face verification.
- Prediction of Wind Power using Supervised Machine Learning Methods.
- Automated Handwritten document Images Recognition, and more...

Course Responsible

- Dr. Huseyin KUSETOGULLARI

E-mail: huseyin.kusetogullari@bth.se

Other Teachers and Course Assistant

- Other Teachers
  - Dr. Majed Elwardy(majed.elwardy@bth.se)
  - Thi My Chinh Chu (thi.my.chinh.chu@bth.se)
  - Guest Lecturers (if possible)

- Course Assistant
  - Thi My Chinh Chu (thi.my.chinh.chu@bth.se)

Course syllabus

- The course syllabus is available at https://canvas.bth.se

- Please check the syllabus in detail including
  - learning outcomes
  - description of the course
  - assessment part and credit point
  - exercises
  - written examination
  - main book and other materials

- Once you complete the course, you are expected to have general and practical knowledge of Applied AI course and implementation of several AI methods to resolve optimization and data science problems.

Assignments and Examination

- There are;
  - 3 Assignments for DV2618 (Assignment 1, 2 and 3)
  - 2 Assignments for DV2659 (Assignment 1 and 3)
- You must include
  - your implementation (if required, Python software)
  - a report (if required)

Three - Weeks Rule

- Accessing the Canvas course webpage is ok.
- Inactive students will be removed from the course.

Submission Deadlines

- Assignment 1 – 20th September, until 23:59
- Assignment 2 – 4th October, until 23:59

- Assignment 3 – 16th October, until 23:59

- *Assignments are individual and must be completed individually.

About Lectures

- There are
  - 10 lectures
  - 3 Labs

About Final Exam

- The date will be announced later.
- It will be a computer-based exam on campus.

Lectures

Lectures

The course provides a historical overview of the development of the AI field, emphasizing major milestones from an application-oriented perspective.

Lecture 1 : INTRODUCTION - Overview of AI, History, Evolution and Applications

Lecture 2 : INTELLIGENT AGENTS, KNOWLEDGE REPRESENTATION IN AI **(Please read sections 2, 7, 8, and 12 from the Book "Artificial Intelligence: A Modern Approach.")**

Lecture 3 : GRAPHS AND SEARCH IN AI **(Please read sections 3 and 4 from the Book "Artificial Intelligence: A Modern Approach.")**

Lecture 4: OPTIMIZATION AND GENETIC ALGORITHM **(Please read Section 3.5.2 and 4.1.4 in the Artificial Intelligence Modern Approach book, Sections 7.1, 7.2, 7.3, 7.4, 7.5, and 7.6 in the Artificial Intelligence (A guide to Intelligent Systems) Book.)**

Lecture 5: DATA MINING AND KNOWLEDGE DISCOVERY **(Please read Section 9 in the Artificial Intelligence (A guide to Intelligent Systems) Book.)**

Lecture 6: MACHINE LEARNING: FUNDAMENTALS OF SUPERVISED, UNSUPERVISED LEARNING **(Please read Sections 18, 19, 20, and 21 in the Artificial Intelligence Modern Approach book.)**

Lecture 7: DEEP LEARNING **(Please read Section 6 in the Artificial Intelligence (A guide to Intelligent Systems) and section 18.7 in theBook and Artificial Intelligence A Modern Approach.)**

Lecture 8: KEY APPLICATIONS: VISION & NATURAL LANGUAGE PROCESSING **(Please read Sections 22 and 24 in the Artificial Intelligence Modern Approach book.)**

Lecture 9: THE ETHICS OF AI, FUTURE OF AI, BENEFITS, CHALLENGES

Lecture 10: GENERATIVE AI, LARGE LANGUAGE MODELS and Finalizing the course

Labs-Exercises

- There will be three different Lab exercises:
  - First one is to use Genetic Algorithm for an optimization problem.
  - Second one is to apply Machine Learning Approaches for Regression and Clustering.
  - Third one is to apply Deep Learning and Machine Learning Methods for Image Classification.
- We use Python software.
- The labs will be hybrid and not mandatory to attend.
- You can bring your own laptop if you join the labs on campus.

Purpose of this course

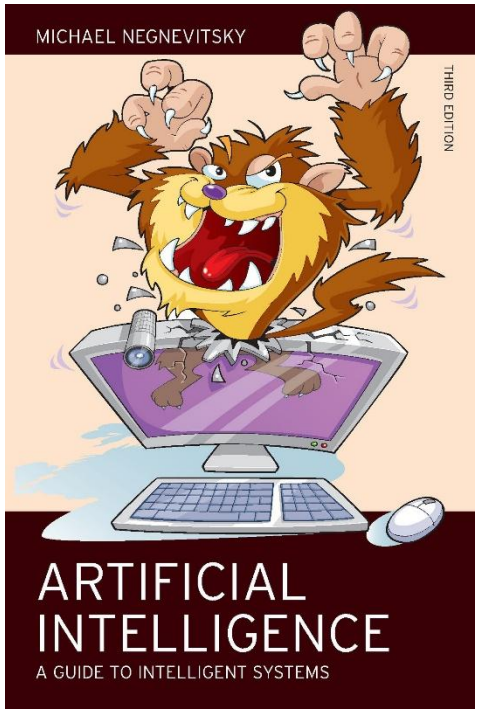
- Learn how to use the basic search methods;
- Understand the basic methods for problem formulation and knowledge representation;
- Understand the usage of automatic reasoning;
- Know some basic concepts related to optimization, data science, prediction, pattern recognition and machine learning.

Book

Artificial Intelligence: A Guide to Intelligent Systems 3rd Edition,

Written by Michael Negnevitsky

https://www.amazon.com/Artificial-Intelligence-Guide-Intelligent-Systems-ebook/dp/B00IZ0806S

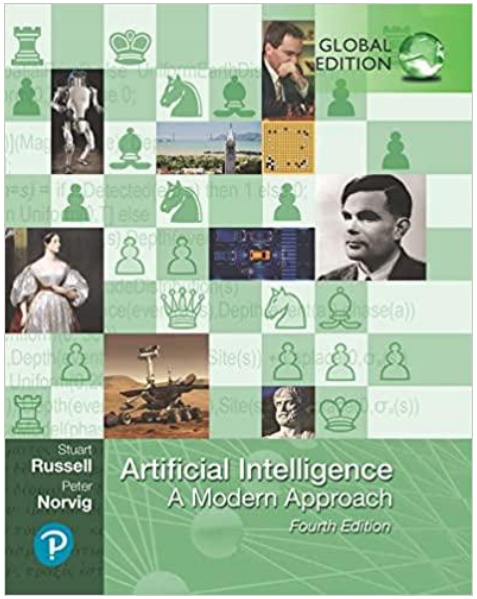


Book-2

Artificial Intelligence: A Modern Approach, Global Edition

Written by Peter Norvig, Stuart Russell

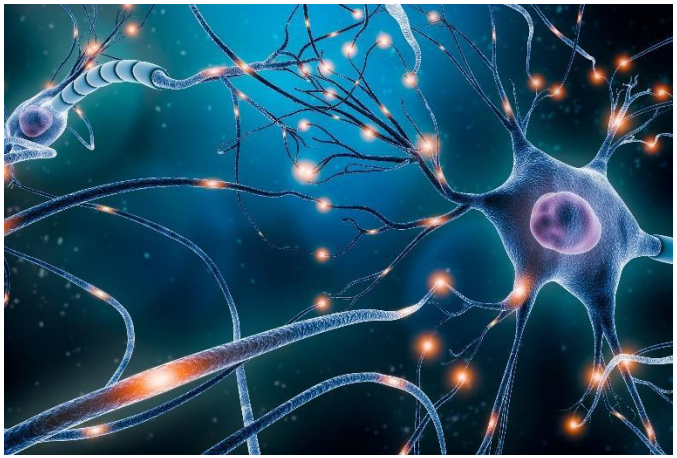
https://www.amazon.se/Artificial-Intelligence-Modern-Approach-Global/dp/1292401133/ref=asc_df_1292401133/?tag=shpngadsglede-21&linkCode=df0&hvadid=476412769423&hvpos=&hvnetw=g&hvrand=8314551043088531213&hvpone=&hvptwo=&hvqmt=&hvdev=c&hvdvcmdl=&hvlocint=&hvlocphy=1012397&hvtargid=pla-1219002048885&psc=1



Recommended research papers

- You can find recommended research papers on the course webpage in Canvas.

Intelligence





The Brain’s Role in Processing and Decision-Making

Emotions and Social Intelligence

Learning and Memory

Creativity and Problem-Solving

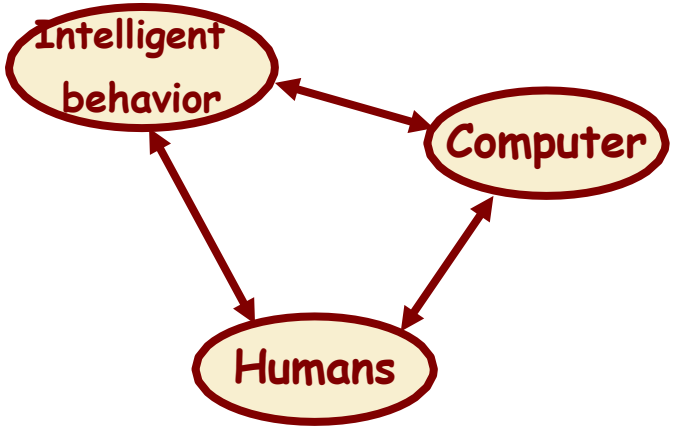
The human brain contains approximately **86 billion neurons [1]**. These neurons are the brain's primary cells, responsible for transmitting information through electrical and chemical signals.

[1]https://pmc.ncbi.nlm.nih.gov/articles/PMC2776484/

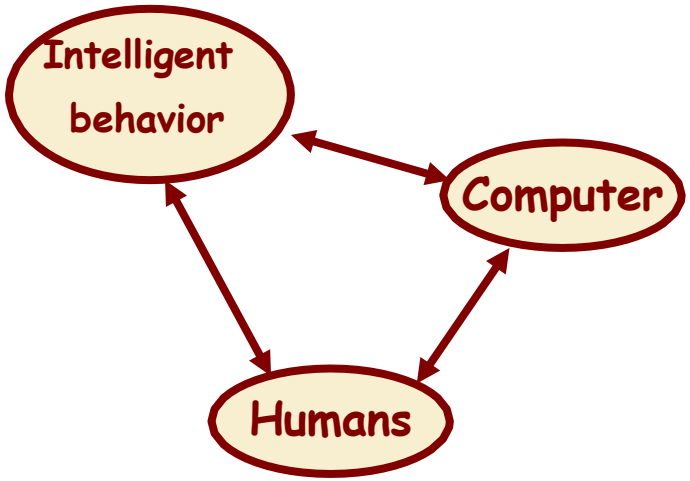
What is Artificial Intelligence?



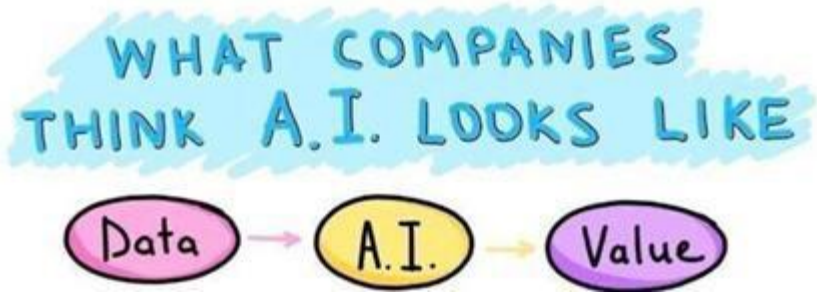
- is a simulation of human intelligence processes by machines, especially computer systems.
- AI refers to the field of computer science dedicated to creating systems or machines that can perform tasks typically requiring human intelligence.
- is defined as the ability of a digital computer or computer-controlled robot to perform tasks commonly associated with human intelligent beings.
- is the intelligence of a machine or computer that enables it to imitate or mimic human capabilities.
- Creating computer systems that perform tasks regarded as requiring intelligence when done by humans.

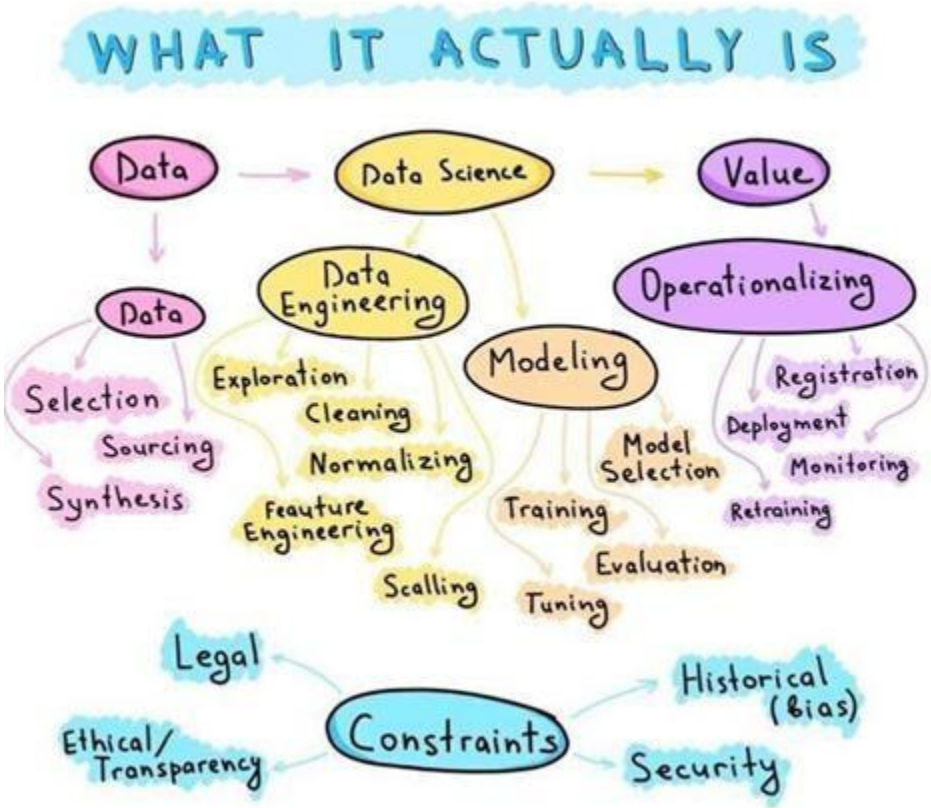




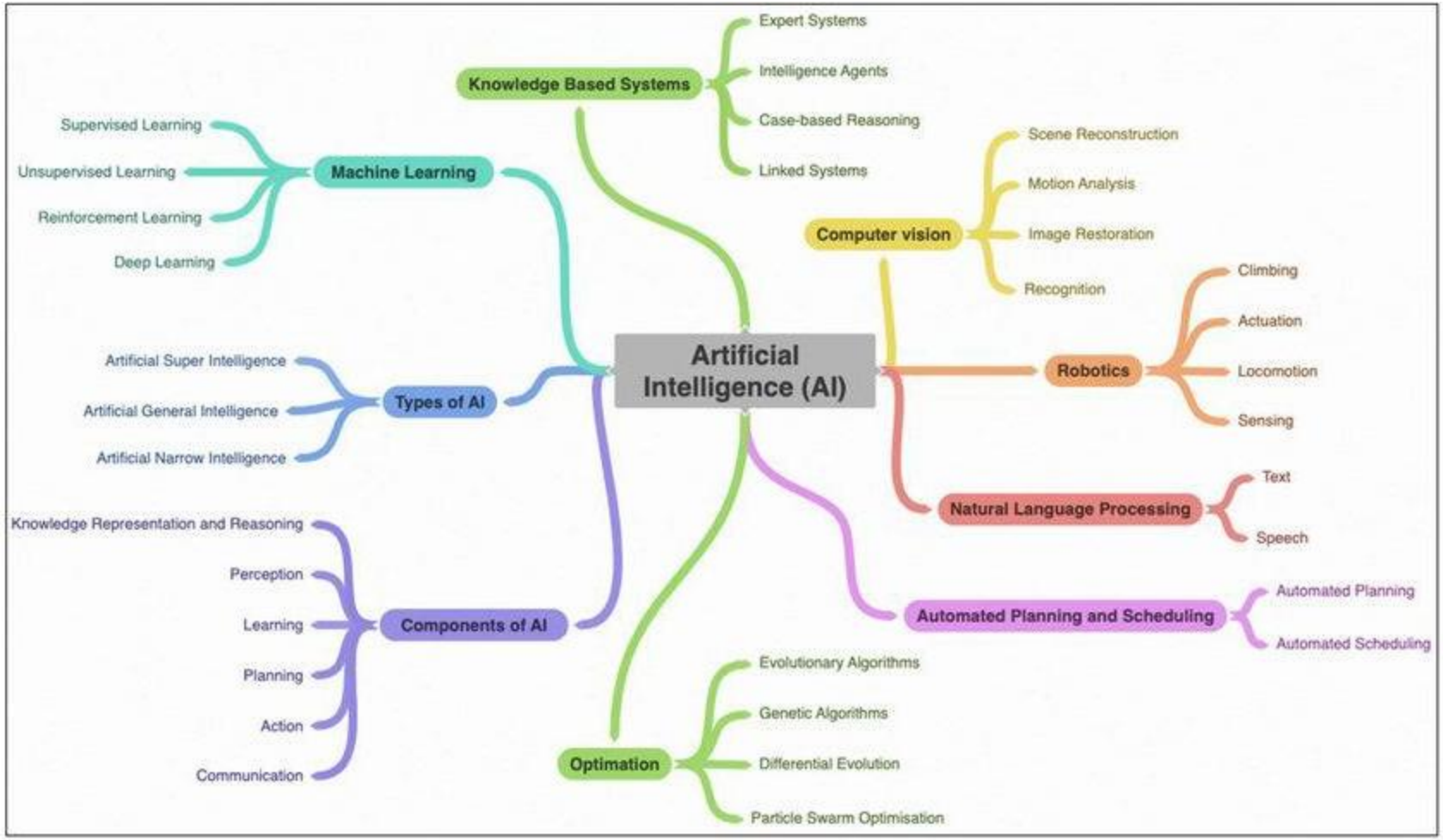

Intelligent behavior
Computer
Humans

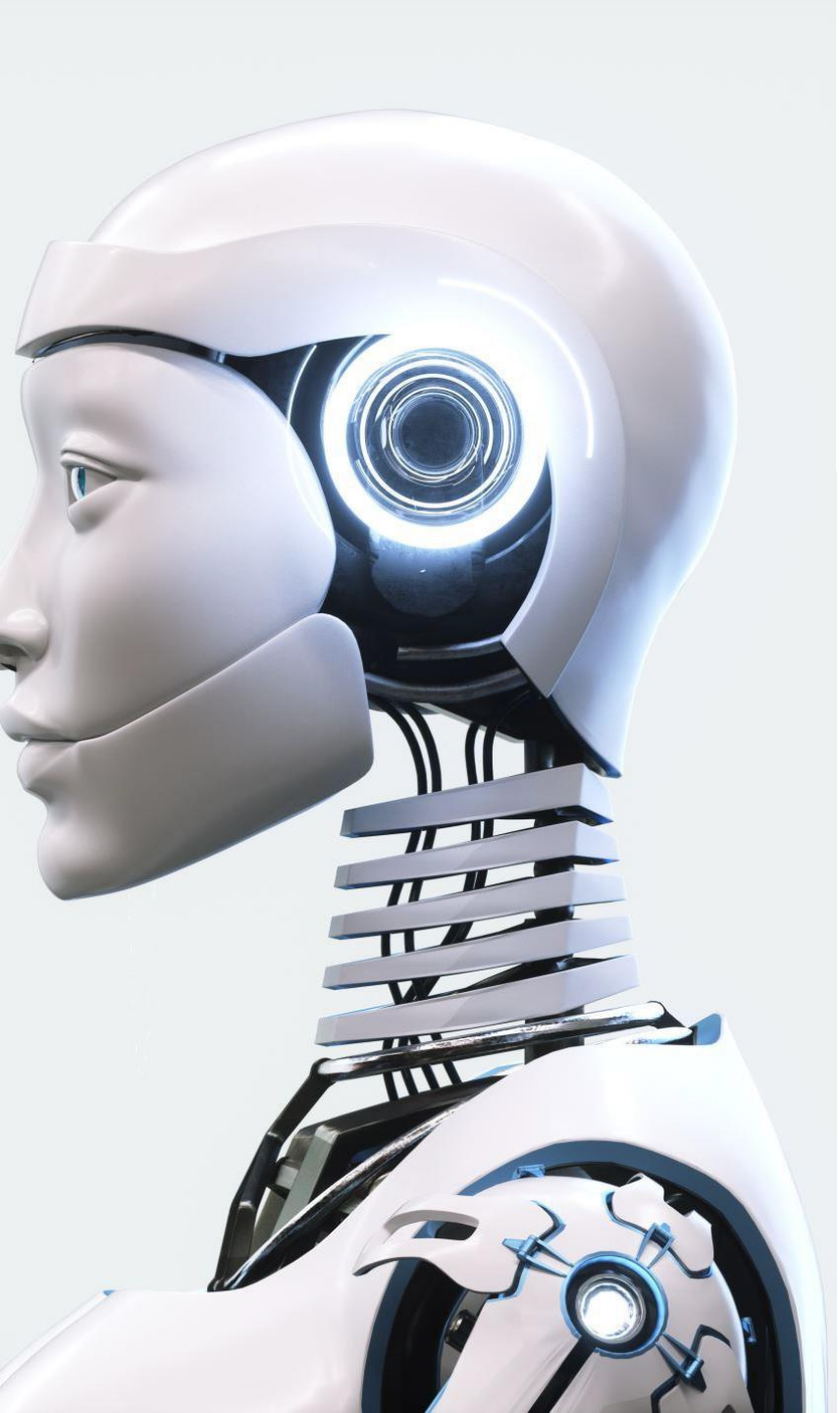
What do people usually think about AI?





Domain


Artificial Intelligence (AI)
Knowledge Based Systems
Expert Systems
Intelligence Agents
Case-based Reasoning
Linked Systems
Machine Learning
Supervised Learning
Unsupervised Learning
Reinforcement Learning
Deep Learning
Types of AI
Artificial Super Intelligence
Artificial General Intelligence
Artificial Narrow Intelligence
Components of AI
Knowledge Representation and Reasoning
Perception
Learning
Planning
Action
Communication
Computer vision
Scene Reconstruction
Motion Analysis
Image Restoration
Recognition
Robotics
Climbing
Actuation
Locomotion
Sensing
Natural Language Processing
Text
Speech
Automated Planning and Scheduling
Automated Planning
Automated Scheduling
Optimation
Evolutionary Algorithms
Genetic Algorithms
Differential Evolution
Particle Swarm Optimisation



Basics of Artificial Intelligence

What kind of AI system do you want to design?

Basics of Artificial Intelligence

- Views of AI fall into four categories (System):

Acting Humanly	Acting Rationally
Thinking Humanly	Thinking Rationally

* Systems that think like humans.
* Systems that act like humans.
* Systems that think rationally.
* Systems that act rationally.

Thinking / Acting | Humanly/Rationally

- Acting humanly — the Turing Test approach.
- Thinking humanly — cognitive modeling. Systems should solve problems the same way humans do.
- Thinking rationally — the use of logic. Need to worry about modeling uncertainty and dealing with complexity.
- Acting rationally — the study of rational agents: agents that maximize the expected value of their performance measure given what they currently know.

Acting Humanly

- How do you distinguish intelligent behavior from intelligence?
- ***Turing test***, by A. Turing, 1950: determining if a program qualifies as artificially intelligent by subjecting it to an interrogation along with a human counterpart.
- The program passes the test if a human judge cannot distinguish between the answers of the program and the answers of the human subject.
- Some say it has passed, some say it hasn't been passed yet.
- http://www.loebner.net/Prizef/loebner-prize.html

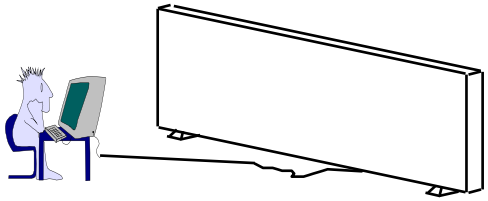
Turing's Test

Turing's Test

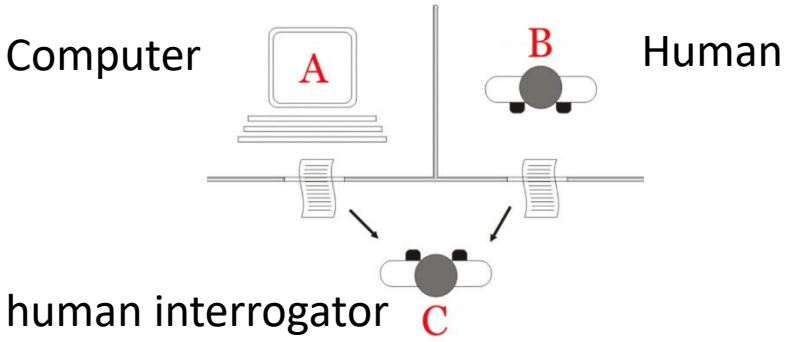
- Test proposed by Alan Turing in 1950
- The computer is asked questions by a human interrogator. It passes the test if the interrogator cannot tell whether the responses come from a person
- No physical interaction

https://academic.oup.com/mind/article/LIX/236/433/986238

Systems that act like humans



?



- You enter a room which has a computer terminal. You have a fixed period of time to type what you want into the terminal, and study the replies. At the other end of the line is either a human being or a computer system.

- If it is a computer system, and at the end of the period you cannot reliably determine whether it is a system or a human, then the system is deemed to be intelligent.

Cont.

- The Turing Test approach
  - a human questioner cannot tell if
    - there is a computer or a human answering his question, via teletype (remote communication)
  - The computer must behave intelligently
- Intelligent behavior
  - to achieve human-level performance in all cognitive tasks

Turing’s Test in Real life (by Google)



Cont.

A computer would need to possess the following minimum capabilities to pass this test:

- A. Natural Language Processing: To enable it to communicate successfully
- B. Knowledge Representation: To store what it knows or hears
- C. Automated Reasoning: To use stored data to answer questions and draw new conclusions.
- D. Machine Learning: To adapt to new circumstances, detect patterns.

The total Turing Test

- Includes two more issues:
  - *Computer vision*
    - to perceive objects (seeing)
  - *Robotics*
    - to move objects (acting)

Thinking Humanly

- "Thinking humanly" in AI involves designing systems that emulate human cognitive processes.
- **Thinking Humanly** means designing an AI system that tries to **think in a way similar to how humans think**.
- This means creating AI that can perform tasks similar to how humans think, reason, and understand.

The main question is:

“How do humans think, learn, remember, and solve problems—and can we make a computer do something similar?”

- This approach is closely related to **cognitive science**, psychology, and neuroscience.

Simple example

Imagine you see this sequence:

2, 4, 6, 8, ?

A human might think: ?

“The numbers increase by 2 each time, so the answer should be 10.”

In the **Thinking Humanly** approach, researchers are interested not only in getting the answer **10**, but also in **how the reasoning process leads to that answer**.

Why is it important?

Thinking Humanly is useful because understanding **human intelligence can inspire artificial intelligence**. Researchers study human **learning, memory, perception, reasoning, creativity, and decision-making** and then try to develop computational models that reproduce some of these abilities.

Thinking Rationally (“laws of thought”)

- **Thinking rationally** means designing an AI system that reasons logically and reaches conclusions by following formal rules of logic.

- "Thinking rationally" in AI refers to designing systems that make decisions or take actions that are logically consistent with their goals, given the available information.

- This means the AI is focused on optimizing outcomes based on a set of rules or principles, ensuring that its behavior aligns with the best possible course of action in a given situation.

The idea comes from the classical **“laws of thought”**: if the AI knows certain facts and rules, it should be able to logically derive a correct conclusion.

Suppose an AI knows:

All humans are mortal.

Socrates is a human.

Using logical reasoning:

Human(Socrates) → Mortal(Socrates)

Therefore:

Socrates is mortal.

The AI reaches the conclusion by applying a logical rule rather than learning it from examples.

AI example: Medical expert system

An AI could have rules such as:

IF a patient has a fever AND a sore throat,
THEN consider a throat infection.

If the system observes:

Fever = True
Sore throat = True

Acting rationally: rational agent

- **Acting rationally** means designing an AI system that **chooses the best possible action to achieve its goals**, based on what it knows and observes.

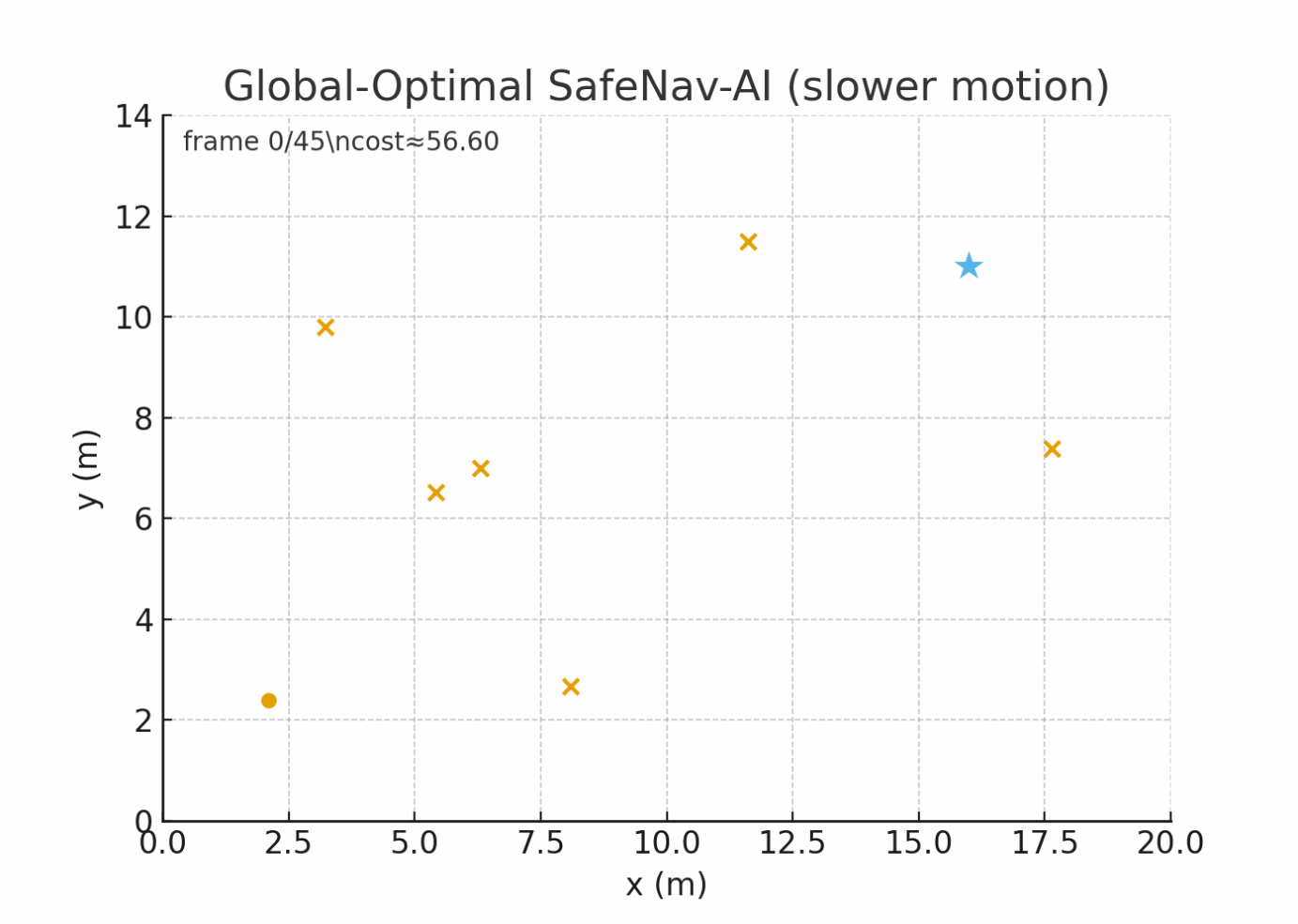
- Do the right thing: that which is expected to maximize goal achievement, given the available information

- Doesn't necessarily involve thinking – e.g., blinking reflex – but thinking should be in the service of rational action

Cont.

- For example, consider a **self-driving car**.
- **Perception:** Cameras and sensors detect pedestrians, cars, traffic lights, and road conditions.
- **Goal:** Reach the destination safely and efficiently.
- **Possible actions:** Accelerate, brake, turn, change lanes.
- **Rational action:** If a pedestrian suddenly enters the road, the rational action is to **brake**, because this best supports the goal of safe driving

Demonstration



Example:

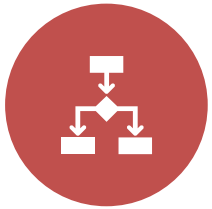
Can we design an AI system that both "thinks humanly" and "behaves rationally?

- Yes, you can design an AI system that "thinks humanly" by mimicking human cognitive processes while also ensuring it "behaves rationally" by making decisions that are logical and goal-oriented.

- **Healthcare** :
- **Thinking Humanly**: The AI mimics how a doctor would analyze symptoms, medical history, and test results by using a knowledge base and reasoning processes similar to a human expert.
- **Behaving Rationally**: It then selects the most effective treatment plan based on the data, optimizing for patient outcomes, and ensuring that the decision is the best possible given the available information.

- **Game**
- **Thinking Humanly:** The AI simulates how a human player would think during the game. For instance, it might consider past experiences (like previous matches or skirmishes) and adapt its strategy based on the opponent's behavior, much like how a human player learns and evolves their gameplay.
- **Behaving Rationally:** The AI evaluates different strategies using **logical reasoning** to choose the one with the highest likelihood of success. For example, it might decide whether to launch an attack or fortify defenses based on a rational assessment of its current strength compared to the opponent. This rational behavior is based on the **goal of maximizing efficiency**, something a human might not always do perfectly due to distractions or errors.

Commons: **acting rationally**, **acting humanly**, **thinking humanly**, and **thinking rationally** in AI



DECISION-MAKING AND PROBLEM-SOLVING



GOAL-ORIENTED BEHAVIOR



ADAPTATION AND LEARNING:



USE OF KNOWLEDGE AND INFORMATION



EVALUATION OF OUTCOMES



ETHICS AND FAIRNESS CONSIDERATIONS

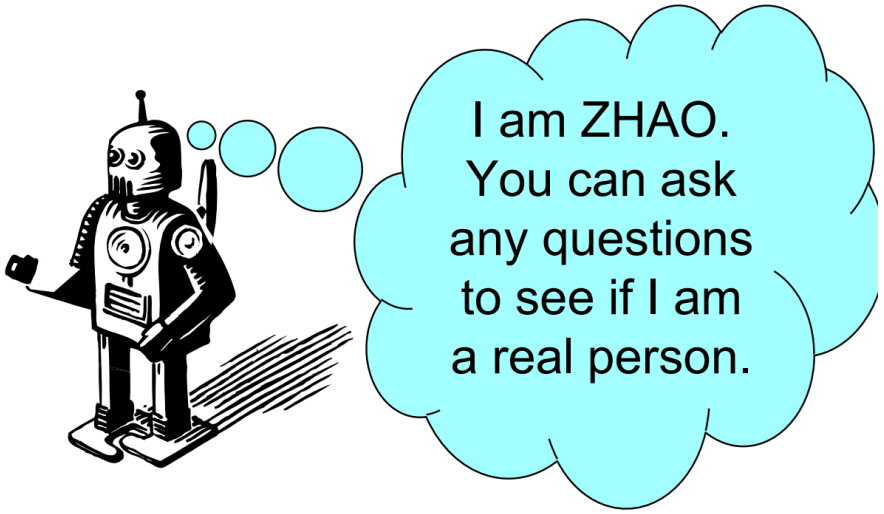
Overview, History and Evaluation

Early work (Around 1800 - 1900）

- Representatives
  - George Boole
  - Alfred North Whitehead
  - Bertrand A. W. Russell
- Main contributions
  - Boolean algebra
  - Principia Mathematica

Early work (1930～）

- Representatives
  - Alan Turing
  - Claude Shannon
  - John von Neumann
- Main contributions
  - Theory of computation, Turing Machine
  - Turing test (to distinguish machine from human)
  - Information theory, application of Boolean algebra
  - von Neumann model of computing machines



AI (1980～)

- Representatives
  - David Rumelhart
  - Lotfi Zdeh
  - John Holland
  - Lawrence Forgel
  - Ingo Rechenberg
  - John Koza
- Main contributions
  - Learning of MLP
  - Fuzzy logic
  - Genetic algorithms
  - Evolutionary programming
  - Evolution strategy
  - Genetic programming

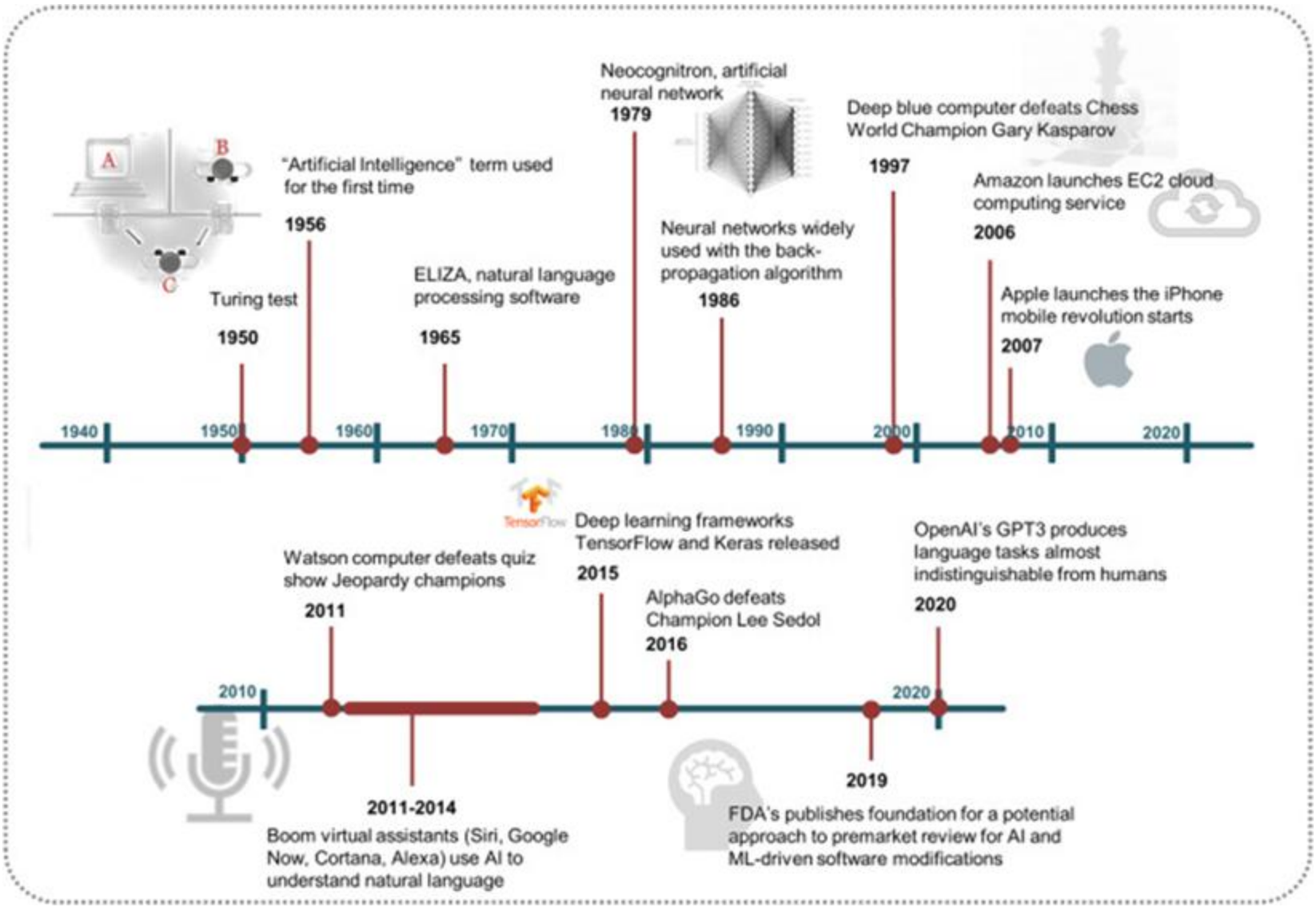
Soft computing
Human like computing and natural computing

AI (2000～)

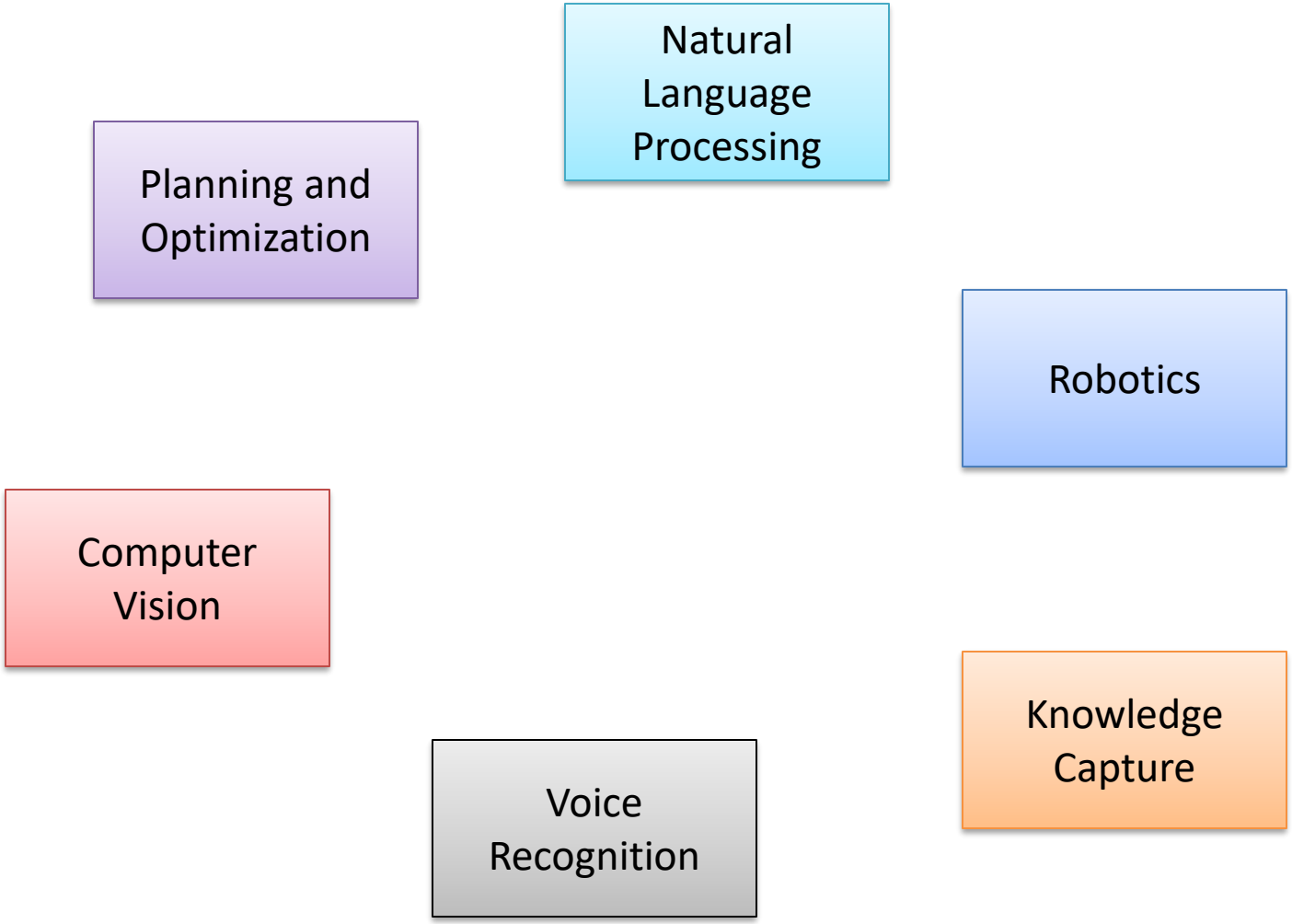
- Representative technologies
  - Internet: Tim Berners-Lee, WWW inventor, 1989
  - Internet of things: Kevin Ashton, MIT Auto-ID Center, 1999
  - Cloud computing
    - Main frame (1950s)
    - Virtual machine (1970s)
    - Cloud (1990s)
  - Big data: John R. Masey, SGI, 1998
  - Deep learning: Geoffrey Hinton, UoT, 2006

A brief summary

- Early work
  - Theoretic foundations
- The second period
  - Acquire knowledge from data / experiences
- The third period
  - Advanced level Intelligence methods


A
B
C
"Artificial Intelligence" term used for the first time
1956
Turing test
1950
ELIZA, natural language processing software
1965
Neocognitron, artificial neural network
1979
Neural networks widely used with the back-propagation algorithm
1986
Deep blue computer defeats Chess World Champion Gary Kasparov
1997
Amazon launches EC2 cloud computing service
2006
Apple launches the iPhone mobile revolution starts
2007
1940
1950
1960
1970
1980
1990
2000
2010
2020
Deep learning frameworks TensorFlow and Keras released
2015
Watson computer defeats quiz show Jeopardy champions
2011
AlphaGo defeats Champion Lee Sedol
2016
OpenAI's GPT3 produces language tasks almost indistinguishable from humans
2020
2010
2020
2019
2011-2014
Boom virtual assistants (Siri, Google Now, Cortana, Alexa) use AI to understand natural language
FDA's publishes foundation for a potential approach to premarket review for AI and ML-driven software modifications

Applications of AI


Natural Language Processing
Planning and Optimization
Robotics
Computer Vision
Knowledge Capture
Voice Recognition

AI in Big Data

- Facebook
- Youtube: Videos
- Instagram: Images
- Twitter: words, sentences etc.

Year	1	2	3	4	5	6	7	8	9	10
2000	pro	mahjong	series	racing	star	world	dance	de	1500	simple
2001	series	simple	pro	x	world	1500	advance	racing	super	hunter
2002	pro	series	advance	simple	world	wars	soccer	star	nba	tennis
2003	series	world	football	battle	pro	club	racing	star	game	dragon
2004	football	club	series	man	world	battle	famicom	mini	classic	final
2005	world	man	wars	racing	series	collection	king	star	battle	x
2006	world	battle	ds	racing	series	dance	fifa	game	portable	ultimate
2007	ds	game	world	battle	series	monster	quest	dance	war	man
2008	world	ds	game	wars	party	american	episode	star	war	dance
2009	episode	world	wars	game	pocket	god	hero	star	force	vs
2010	world	game	super	episode	pro	dead	star	wars	alien	heroes
2011	3d	game	war	world	adventures	space	super	dead	man	vs
2012	3d	game	hd	legends	super	war	edition	heroes	black	dark
2013	hd	world	dead	dungeon	space	star	war	wars	super	3d
2014	war	dead	putt	world	space	adventure	dark	planet	time	game
2015	life	shadows	war	dark	elysian	fate	night	battlefield	hand	hardline

Categorical Data: words

	A	B	C	D	E	F	G	H	I	J	K	L
1	fixed acid	volatile ac	citric acid	residual si	chlorides	free sulfu	total sulfu	density	pH	sulphates	alcohol	quality
2	7.4	0.7	0	1.9	0.076	11	34	0.9978	3.51	0.56	9.4	5
3	7.8	0.88	0	2.6	0.098	25	67	0.9968	3.2	0.68	9.8	5
4	7.8	0.76	0.04	2.3	0.092	15	54	0.997	3.26	0.65	9.8	5
5	11.2	0.28	0.56	1.9	0.075	17	60	0.998	3.16	0.58	9.8	6
6	7.4	0.7	0	1.9	0.076	11	34	0.9978	3.51	0.56	9.4	5
7	7.4	0.66	0	1.8	0.075	13	40	0.9978	3.51	0.56	9.4	5
8	7.9	0.6	0.06	1.6	0.069	15	59	0.9964	3.3	0.46	9.4	5
9	7.3	0.65	0	1.2	0.065	15	21	0.9946	3.39	0.47	10	7
10	7.8	0.58	0.02	2	0.073	9	18	0.9968	3.36	0.57	9.5	7
11	7.5	0.5	0.36	6.1	0.071	17	102	0.9978	3.35	0.8	10.5	5
12	6.7	0.58	0.08	1.8	0.097	15	65	0.9959	3.28	0.54	9.2	5
13	7.5	0.5	0.36	6.1	0.071	17	102	0.9978	3.35	0.8	10.5	5
14	5.6	0.615	0	1.6	0.089	16	59	0.9943	3.58	0.52	9.9	5
15	7.8	0.61	0.29	1.6	0.114	9	29	0.9974	3.26	1.56	9.1	5
16	8.9	0.62	0.18	3.8	0.176	52	145	0.9986	3.16	0.88	9.2	5
17	8.9	0.62	0.19	3.9	0.17	51	148	0.9986	3.17	0.93	9.2	5
18	8.5	0.28	0.56	1.8	0.092	35	103	0.9969	3.3	0.75	10.5	7
19	8.1	0.56	0.28	1.7	0.368	16	56	0.9968	3.11	1.28	9.3	5
20	7.4	0.59	0.08	4.4	0.086	6	29	0.9974	3.38	0.5	9	4
21	7.9	0.32	0.51	1.8	0.341	17	56	0.9969	3.04	1.08	9.2	6
22	8.9	0.22	0.48	1.8	0.077	29	60	0.9968	3.39	0.53	9.4	6
23	7.6	0.39	0.31	2.3	0.082	23	71	0.9982	3.52	0.65	9.7	5

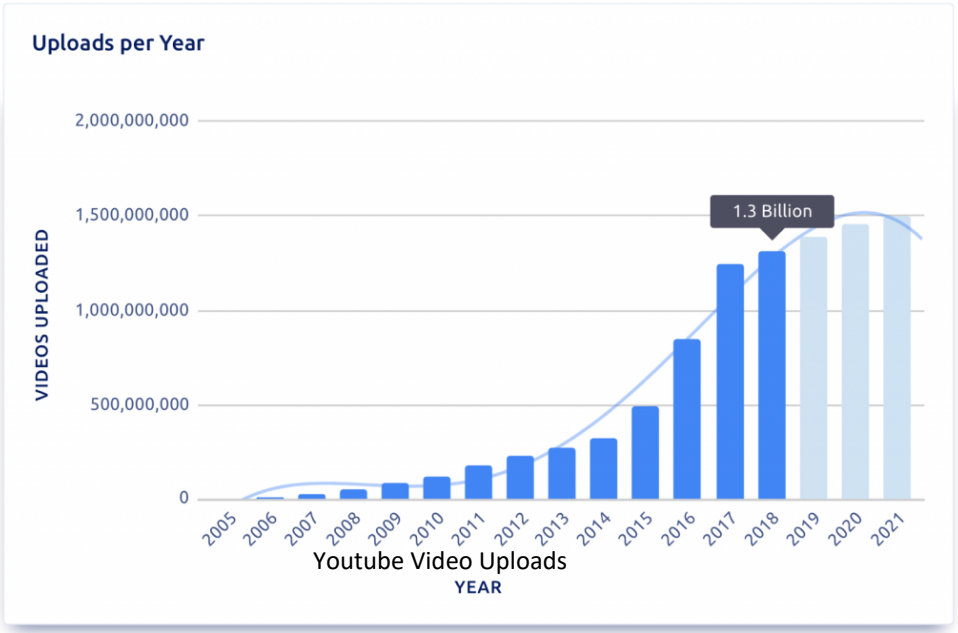
Numerical Data

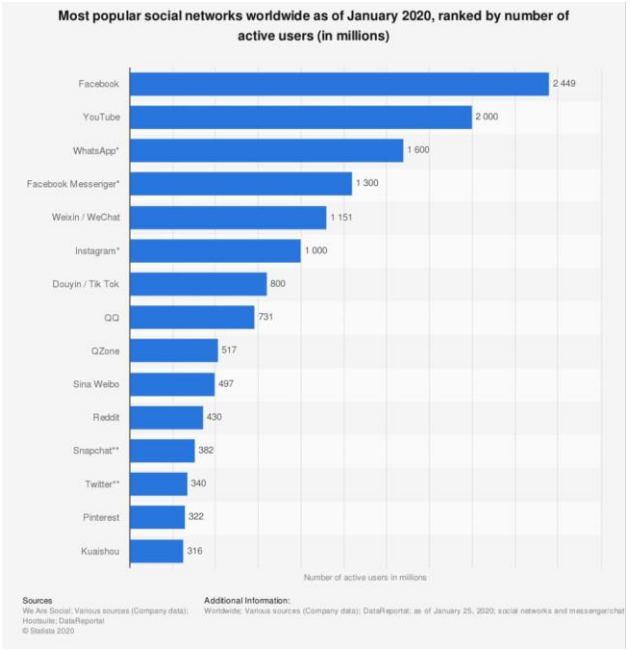


Visual data: images and Videos

Big Data

- There are 2.375bn billion monthly active users. Facebook generates 4 new petabytes (1 petabyte= 1000 TB) of data per day.
- There are 330 million monthly active users and 145 million daily active users on Twitter. 500 million tweets are sent out per day.





Challenges in Big Data

- There are many challenges for the future in big data management and analytics.
  - Pre-processing,
  - Analytics architecture,
  - Evaluation,
  - Imbalance,
  - Anomaly Detection,
  - Developing learning methods,
  - Time-evolving data,
  - Compression,
  - Visualization, etc.

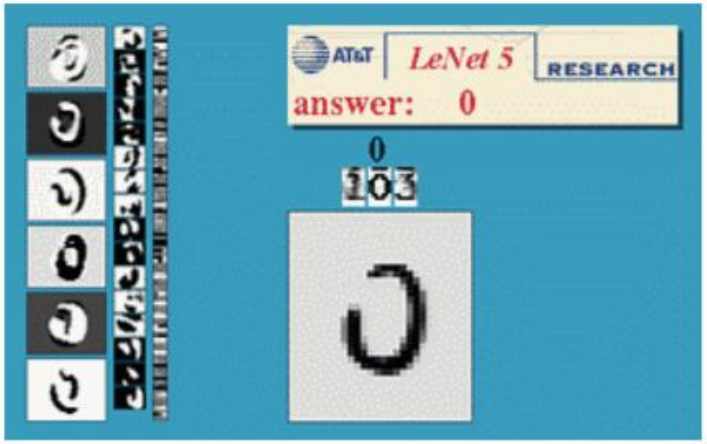
Motivation



- AI Algorithms are needed for tasks which are too complex for humans.
  - Impractical
  - Time consuming
  - Hard to solve
- The size and complexity of these data sets means that humans are unable to extract useful information from them.
- For a large amount of data, an AI algorithm will work out by exploring that data to predict the result.
- Data is everywhere in abundant amounts.
- For instance:
  - Surveillance cameras continuously capture video,
  - Most financial transactions are recorded
  - Satellites generate tera-bytes of data every year
  - Digitized written texts from our libraries etc.
- The main purposes of applying AI in human life are to make easier, safer and better.

Visual Data: Application Areas

Optical character recognition (OCR)



Digit recognition



License plate readers



Biometric Recognition

Ear, eyes, Hand, finger, Gait etc.

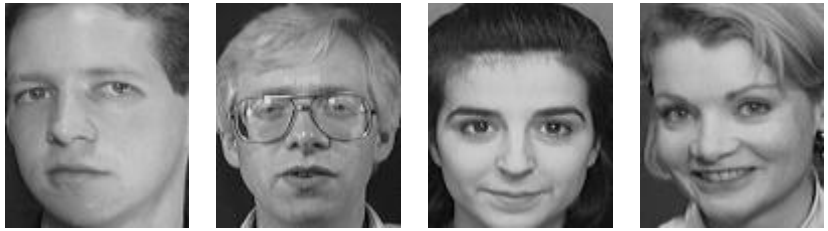
Visual Data: Application Areas

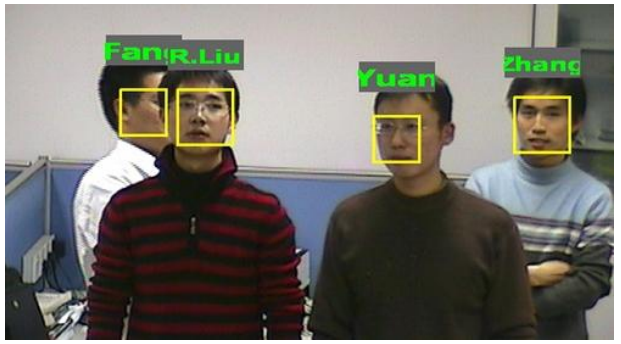
- Face Recognition

Training examples of a person



Test images





Visual Data: Application Areas

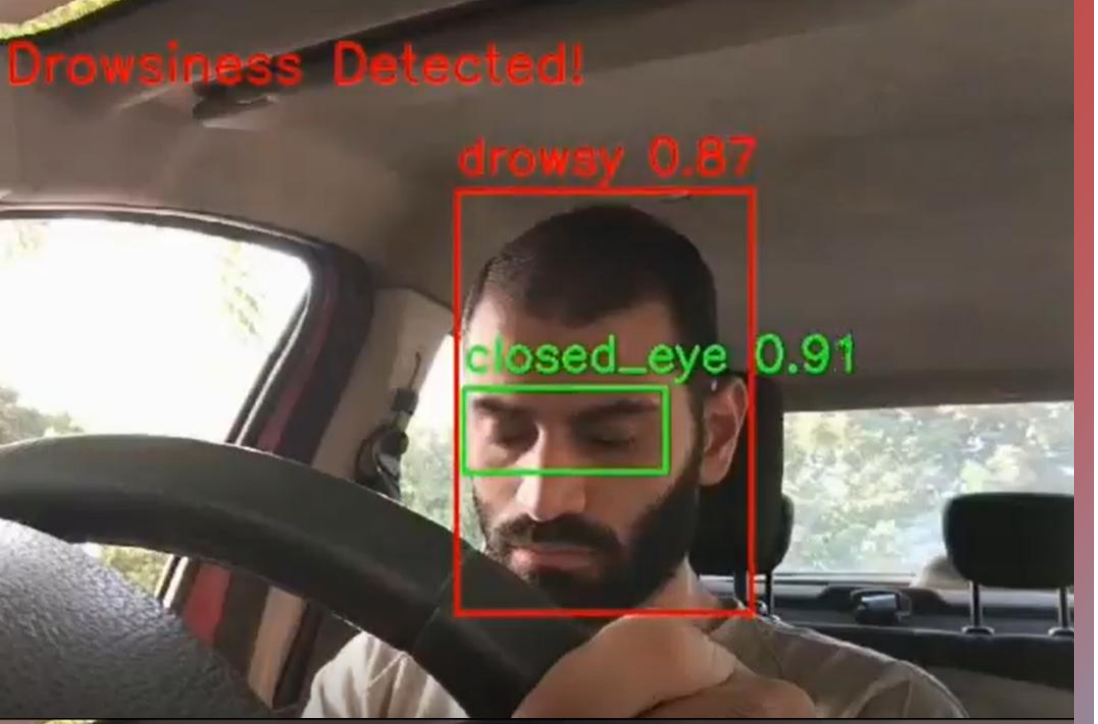
Automotive Industry

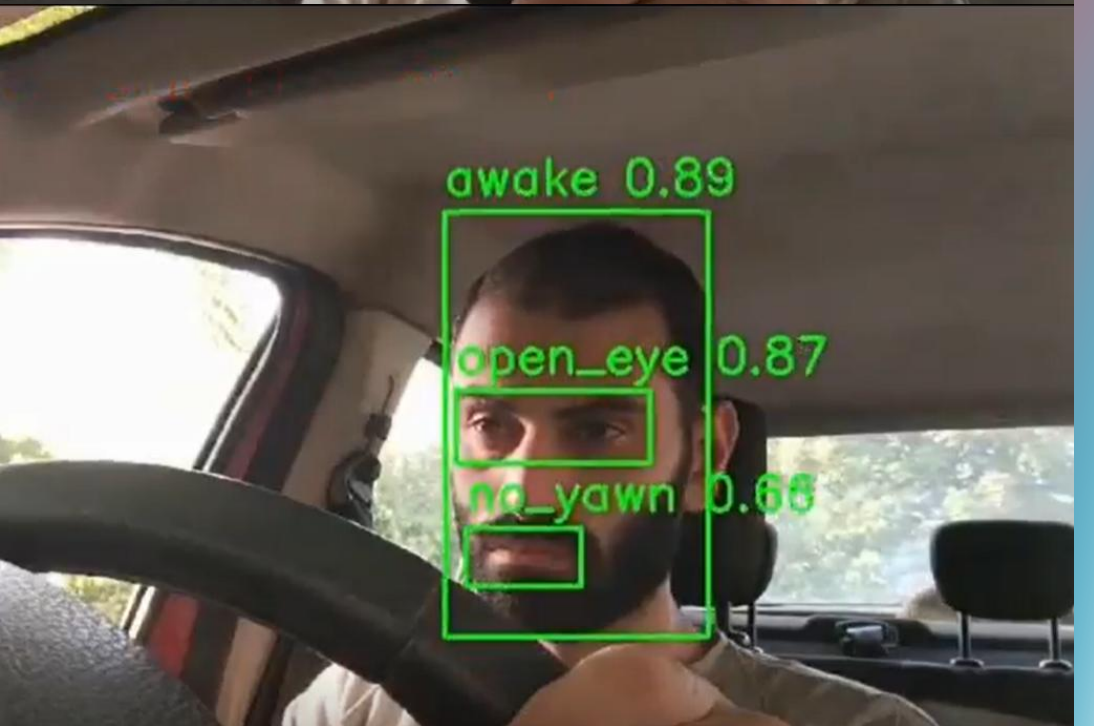


Vision systems in high-end BMW, Tesla, Volvo models
– Object Detection and recognition
– Sign Detection and Recognition
– Pedestrian existence warning
– Existence of other cars warning
– Lane distance warning
– Headway monitoring and warning









Detection and Classification

Problem Definition

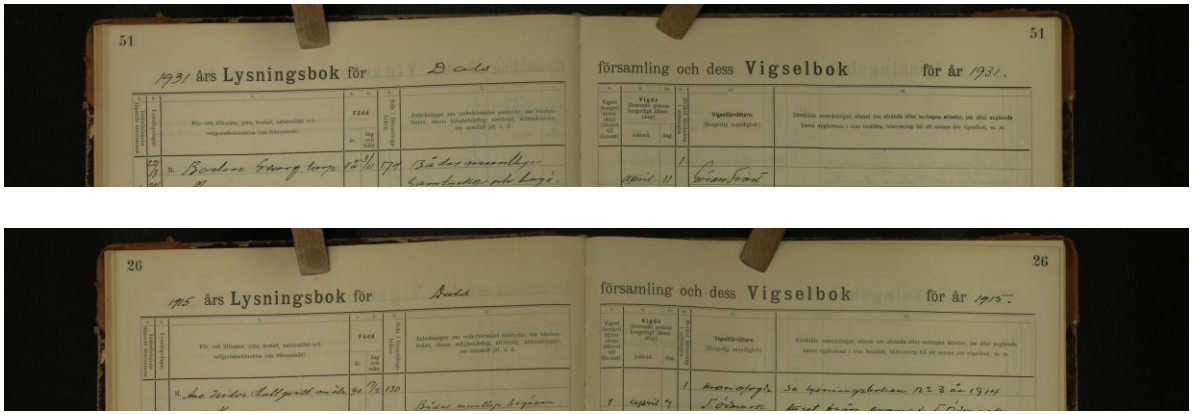
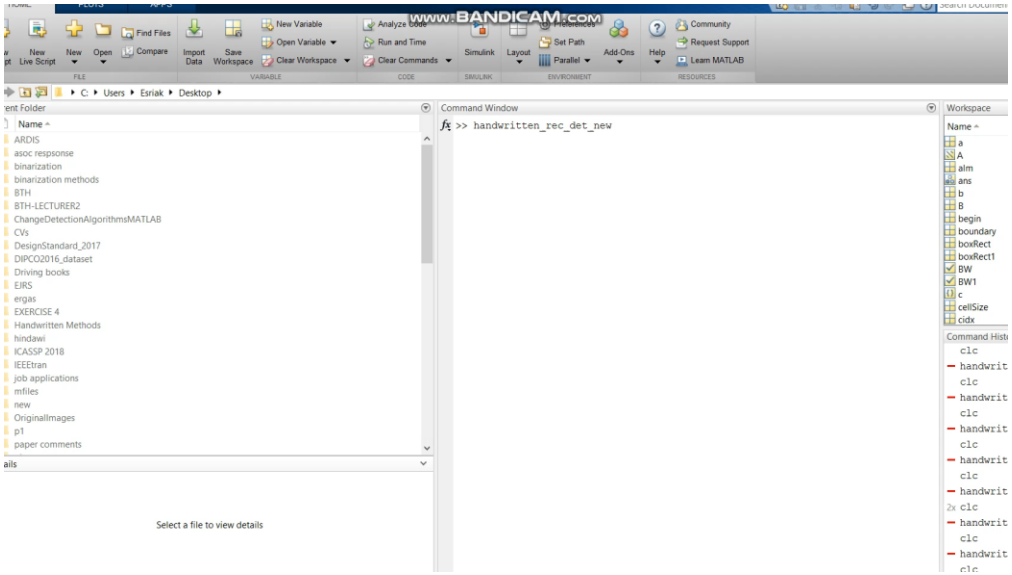


Illustration of part of Document Images

Demonstration



AI in Remote Sensing



Global warming is one of the most important issue over the world.

Finding changes on Earth may help to decrease the affects of the human behaviours.



Change detection is the process of identifying differences between two images acquired in the same geographical area at different times.



This is an important research area that used in many applications

City planning, forestry management,

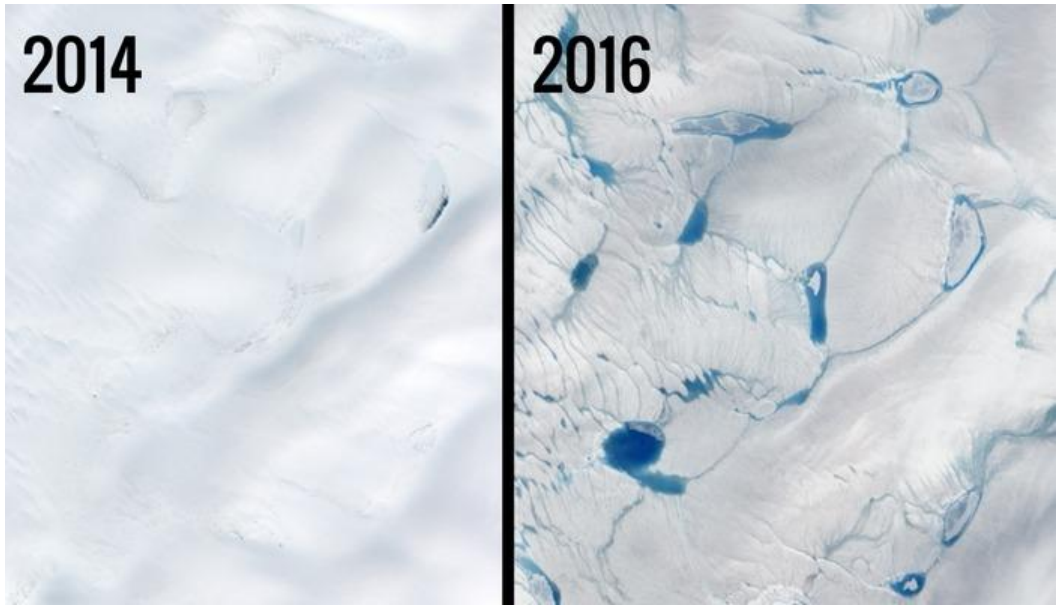
environmental monitoring,

land management,

natural disaster monitoring,

urban planning, and observing the temperature changes in the poles etc.

Cont.

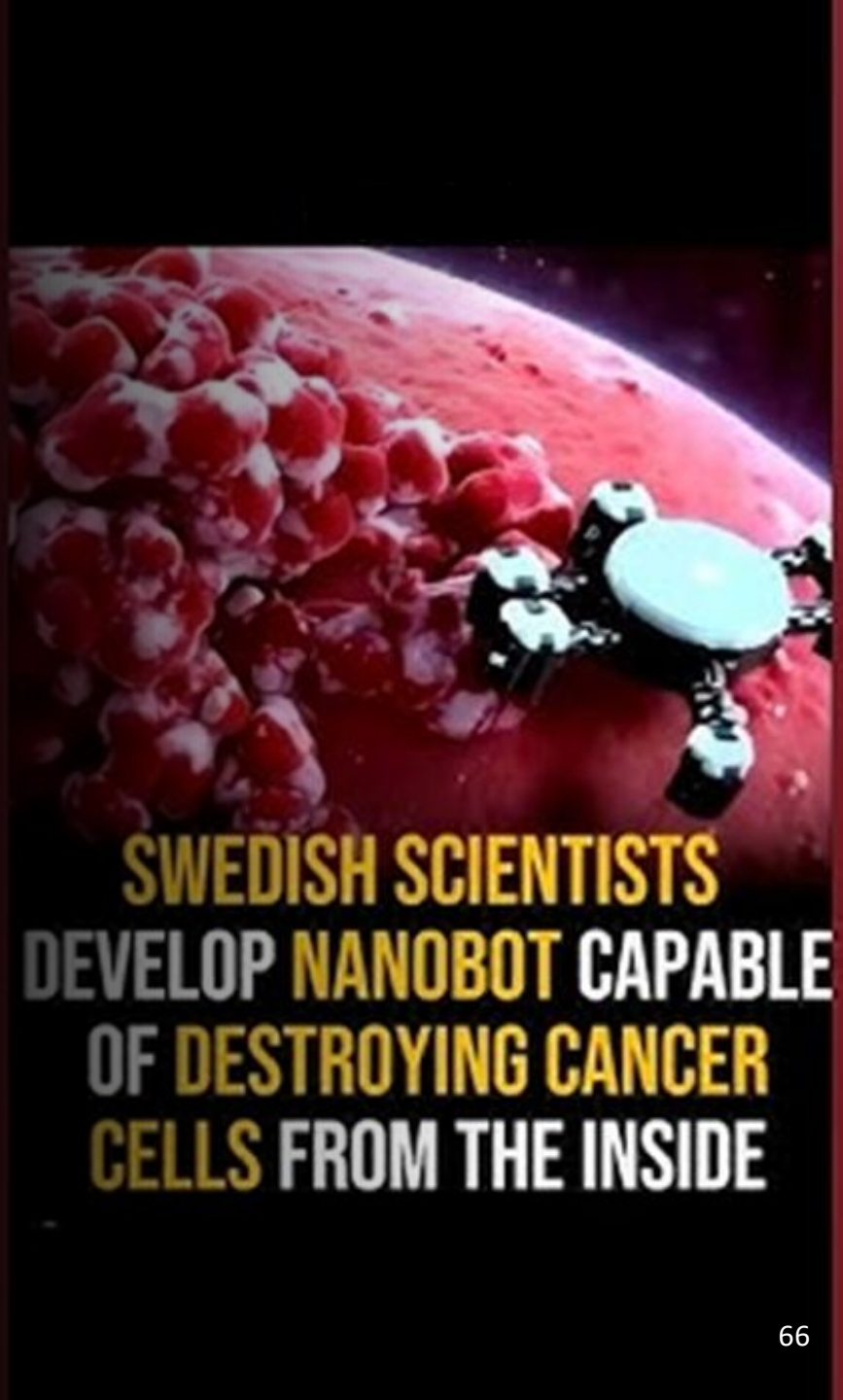


https://earthengine.google.com/timelapse/

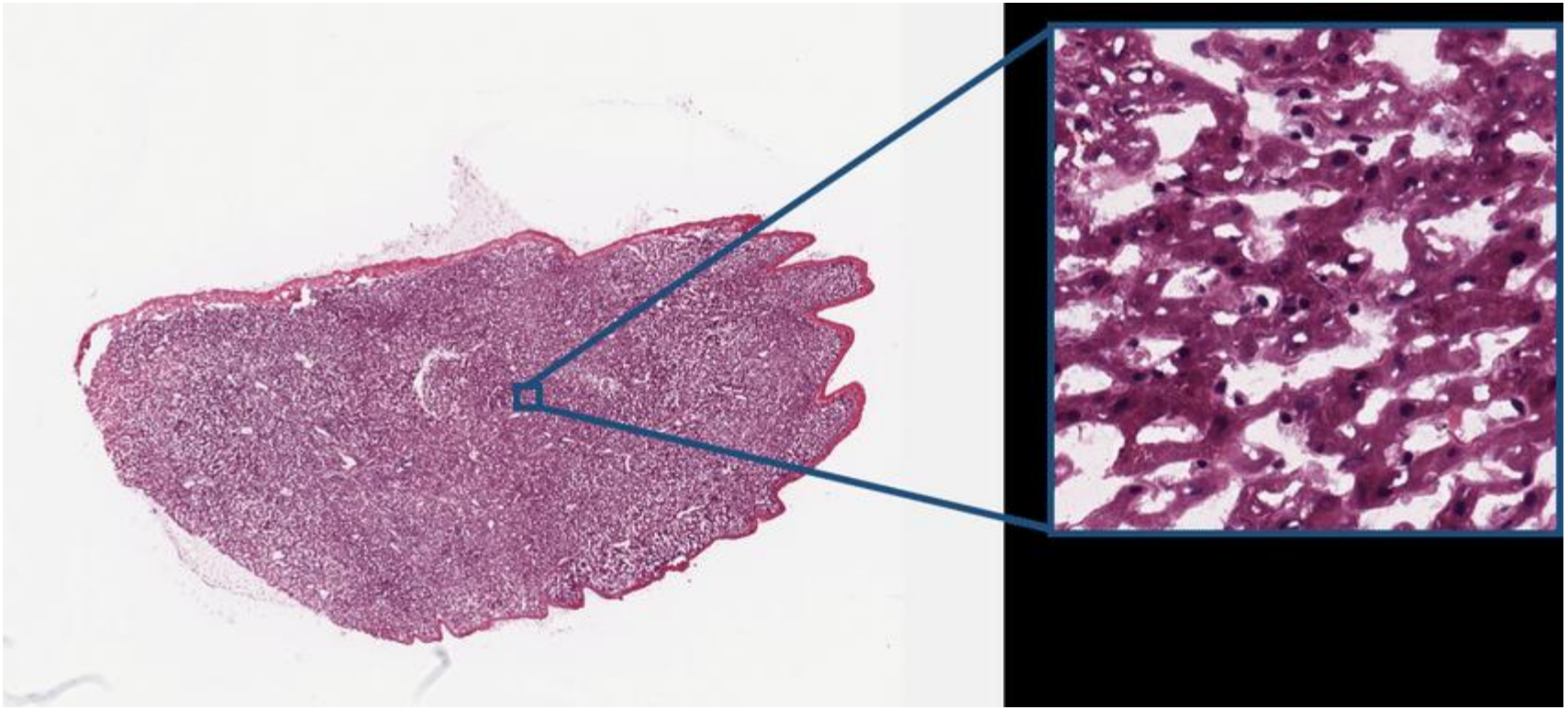
Exceptional Early Ice Melt, Greenland

Medical Image Analysis using AI

Nanobot



https://news.ki.se/nanorobot-with-hidden-weapon-kills-cancer-cells



Whole-slide images generated at diagnostic resolution are very large: a typical whole slide image may contain 100,000x100,000 pixels

AI in Medical Images

- Computer-aided based Breast Cancer diagnosis using Deep Learning methods

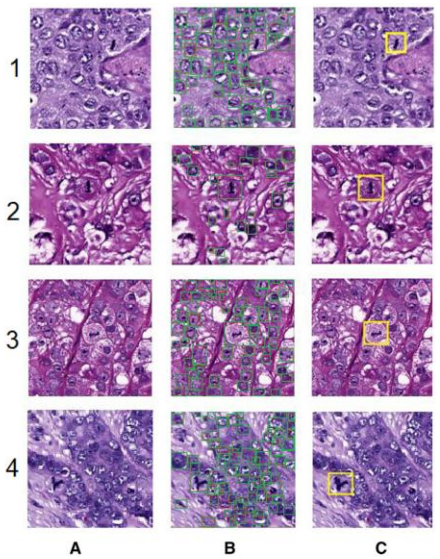


Fig. 9 The qualitative results obtained using the proposed framework, **a** input patches, **b** nuclei detection results, **c** mitosis classification results

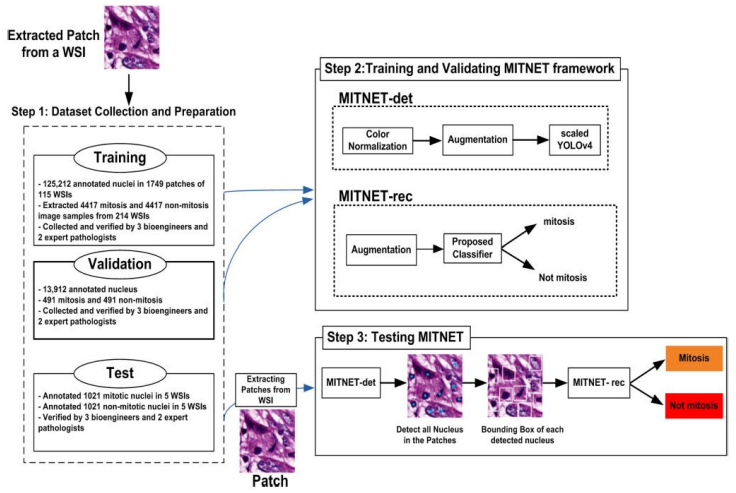
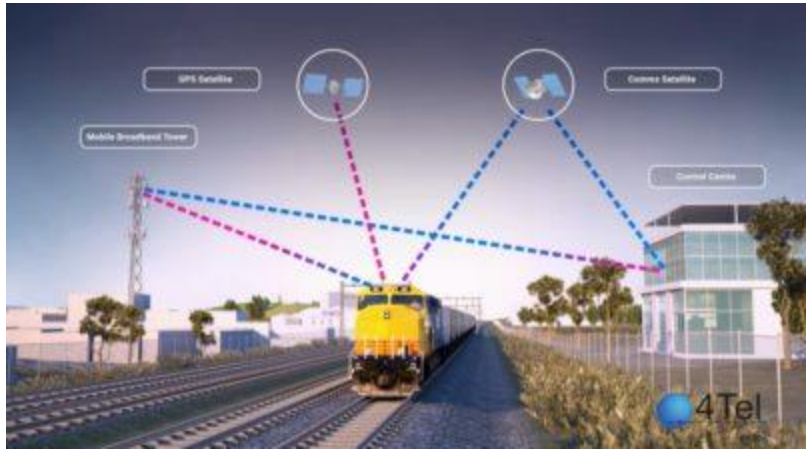


Fig. 5 Flowchart of the proposed methodology and created datasets employed in developing and validating a two-stage deep learning-based method to detect nuclei and recognize mitosis in WSIs

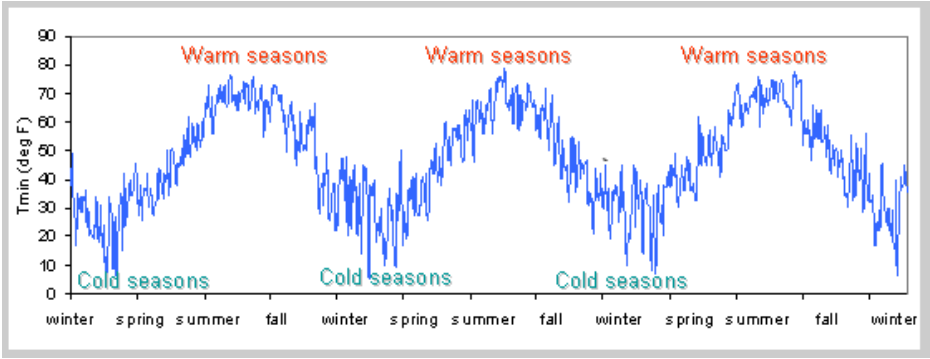
AI in Train Scheduling

- Optimisation of train networks can provide return on all optimality factors;
  - profit,
  - timeliness,
  - robustness.
- Preventing accidents



Time Series Analysis and Application Areas

- A time series is a series of data points indexed (or listed or graphed) in time order.



- Most commonly, a time series is a sequence taken at successive equally spaced points in time.
- It is a sequence of discrete-time data.

AI in Games









More

- **Text Classification:** is the process of assigning categories (tags) to unstructured text data. This essential task of Natural Language Processing (NLP) makes it easy to organize and structure complex text, turning it into meaningful data.
- **Topic Analysis:** helps to understand the main themes or subjects of a text
- **Sentiment Analysis:** consists of analyzing the emotions that underlie any given text. Suppose you are analyzing a series of reviews about your mobile app.
- **Language Detection:** allows you to classify a text based on its language.
- **Keyword Extraction:** Utilizing a keyword extractor allows you to index data to be searched, summarize the content of a text
- **Feature Extraction:** helps identify specific characteristics of a product or service in a set of data.

Generating Data

- Modalities of Generative AI:
  - Text → LLMs (e.g., ChatGPT, Claude).
  - Images → Stable Diffusion, DALL·E, MidJourney.
  - Audio → AI voice cloning (e.g., ElevenLabs) and music generation (e.g., Jukebox).
  - Video → Runway Gen-2, Pika Labs.
  - Multimodal → Models that handle text, images, and other data together (e.g., GPT-4V, Gemini).




OpenAI
ChatGPT

Generate a song using AI ☺

https://aimusic.fm/community/

Do we need AI in our lives?



Enhancing Efficiency and Innovation: In sectors like agriculture, logistics, and energy



Enabling Accessibility and Quality of Life: personalized recommendations, smart home management, and efficient transportation



Supporting Complex Decision-Making

Job Opportunities in AI

- Artificial Intelligence Engineer (AI Engineer)
- Machine Learning Engineer
- Data Scientist
- AI Researcher
- NLP Specialist
- Computer Vision Specialist
- More

Thank you for listening!!!